

主 题:	Invitation to review a manuscript for Scientific Data from Dr He	
发件人:	"Scientific Data" <do-not-reply@springernature.com>	2026-1-10 14:11:44
收件人:	duansh@ustb.edu.cn	

Invitation to review "AirSense-R: A Multimodal Robotic Dataset for Spatially Resolved Indoor Air Quality Monitoring and Navigation Using LiDAR, IMU, and IR Camera Sensing"

Should you accept to review this manuscript, your report would be due within 21 days.

Dear Dr Duan,

We have received a manuscript for Scientific Data that we think falls within your area of expertise. Our reviewers are integral to ensuring we have the highest-quality publication.

We would greatly appreciate it if you could let us know if you are available to review by accepting or declining the invitation link below.

Title: AirSense-R: A Multimodal Robotic Dataset for Spatially Resolved Indoor Air Quality Monitoring and Navigation Using LiDAR, IMU, and IR Camera Sensing

Abstract: Air pollution is one of the most serious concerns indoors, and most low-cost air pollution monitoring systems deployed indoors involve fixed air quality sensors that lack spatial resolution and are incapable of observing variations of air quality in space. To address this issue, we have developed AirSense-R, an air quality monitoring system that is based on a robotic platform and has spatial resolution and autonomy, allowing it to move and explore any environment on its own. The proposed system combines a two-dimensional LiDAR system with a Raspberry Pi for autonomous path tracking, in addition to air quality measurements obtained by means of a nondispersive infrared CO₂ sensor with high precision using a laser PM_{2.5} sensor. In addition to air-quality data, the platform records comprehensive inertial and spatial information, including accelerometer, gyroscope, magnetometer, and LiDAR readings alongside camera data for the fine-tuning of navigation algorithms and for comparative studies between LiDAR- and camera-based navigation approaches. To address privacy concerns in indoor environments, all visual data are captured using an infrared (IR) camera, ensuring anonymity and eliminating identifiable visual information. As the robot moves within an environment, it continuously acquires and synchronizes measurements such that spatially resolved pollutant concentrations can be correlated with room geometry and occupancy conditions. By combining autonomous navigation, multi-sensor data acquisition, and privacy-aware design, the system provides an openly available, multimodal dataset suitable for environmental assessment, ventilation studies, and the development of robust robotic sensing frameworks for indoor air-quality monitoring.

Authors: Amith Khandakar, Mohammed Al-Salahi, Md Masum Mia, Mohamed Arselene Ayari, Zubair Ahmad, Salman Saadat

We hope to hear from you soon.

Kind regards,

Jie He
Editor
Scientific Data

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potential competing interests, so we ask you to draw our attention to anything that might affect your review, and to decline submissions where it may be hard to remain objective.

Contact scientificdata@nature.com if you need any assistance with this request using this submission ID: 8e0584a1-66a8-4025-9118-f2732ed25b26

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